Regulatory Capture via Price Aggregator

Gabriel Cañedo Riedel

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Motivation

Large firms support regulations that seem costly for them:

- Meta: Zuckerberg asked Congress for tighter content-moderation rules.
- Amazon: pushing for doubling the minimum wage.
- Microsoft: backed 'clean grid' computing legislation.
- ExxonMobil: lobbyists pushing for sub-2°C policies.

Why would large firms lobby for regulations that raise their own costs?

Prior Work

When a regulation raises costs more for some firms than others, high-cost firms contract. The reduction in supply drives up the market price — and low-cost incumbents, now facing less competition at a higher price, expand and profit.

All existing formal models of this mechanism work in homogeneous goods markets.

Because all existing models use homogeneous goods, the only available channel is quantities — all firms sell at the same price, so regulation cannot reallocate demand through prices.

In differentiated goods markets, a cost shock can shift relative prices directly, reallocating demand across firms. **This channel has not been formalized.**

This Paper

A model without a relative price channel will attribute all market share reallocation to quantities — exit, contraction, expansion. Any reallocation that operates through relative prices is invisible to it. **Regulation's effects on market structure are therefore systematically underestimated.**

- A firm's **relative** marginal cost can fall even when its own costs rise — if rivals' costs rise by more. Lower relative cost $\Rightarrow$ relative price falls $\Rightarrow$ market share and profit rise.
- **Not all regulations work this way.** I classify cost-increasing regulations into three types: **scale** (neutral), **location** (harms the most productive), **shape** (benefits them).

Outline

1. Literature.
2. Theory: regulation types and main results.
3. Illustration: CES demand with Pareto costs.
4. Application: minimum wage.
5. Conclusion.

Literature

The primary contribution is to the **RRC literature**, extended to a new environment. The application to **minimum wage** produces the first product market mechanism with differentiated goods. Two additional literatures provide context:

- **Regulator as agent (micro/GT).**
  Seminal: Stigler (1971), Peltzman (1976), Grossman and Helpman (1994); Survey: Dal Bó (2006); Recent synthesis/empirics: Bombardini and Trebbi (2020).
- **Raising Rivals' Costs (IO/PE).**
  Canonical: Salop and Scheffman (1983, 1987), Ordover et al. (1990), Hviid and Olczak (2016). Heterogeneous compliance in environmental settings: Kennard (2020).
- **Firm size and capture (applied/macro).**
  Macedoni and Weinberger (2024) (fixed-cost channel). Empirics: Fan and Zhou (2023), Huneeus and Kim (2024), Crane and Koch (2025).
- **Minimum wage and firm outcomes (labor).**
  Rao and Risch (2026); Dustmann et al. (2022); Engbom and Moser (2022); Haelbig et al. (2023); Draca et al. (2011).

→ Detailed literature review.

Raising Rivals' Costs vs Capture via Aggregator

	RRC Literature	**This Paper**
Market structure	Oligopoly with **homogeneous goods**	Monopolistic competition
Strategic interaction	Essential (Cournot/Stackelberg)	Not required
Transmission	Cost shock $\rightarrow$ **quantities**	Cost shock $\rightarrow$ **relative prices**
Entry barriers	Required	Not required

Theoretical Framework

Environment

- Continuum of differentiated goods $\omega \in (0, 1)$, in static closed economy.
- Consumers represented by a homothetic demand system with price aggregator $A(\mathbf{p})$[1]:

$$x_\omega = x_\omega \left(\frac{p_\omega}{A(\mathbf{p})} \right)$$

- Heterogeneous firms in monopolistic competition with marginal cost $\psi_\omega > 0$ maximize:

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega$$

[1] HSA demand systems: Matsuyama and Ushchev (2020).

Relative Marginal Cost

The firm's profit can be rewritten as:

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Where $s_\omega(\cdot)$ is the share function, Y is expenditure, and $A(\cdot)$ a homothetic aggregator. $z_\omega \equiv p_\omega / A(\mathbf{p})$ is the relative price.

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Where $s_\omega(\cdot)$ is the share function, Y is expenditure, and $A(\cdot)$ a homothetic aggregator. $z_\omega \equiv p_\omega / A(\mathbf{p})$ is the relative price.

Key implication: Maximizing over p_ω is equivalent to maximizing over z_ω, and equilibrium profit and market share are decreasing in the **relative marginal cost** $\boldsymbol{\psi_\omega / A(\mathbf{p})}$: $\partial \pi^*_\omega / \partial (\psi_\omega / A(\mathbf{p})) < 0$.

When can higher costs help a firm?

From $\partial\pi^*_\omega/\partial(\psi_\omega/A(\mathbf{p})) < 0$ and $\mathbf{p}^* = \boldsymbol{\mu\psi}$, firm ω benefits from a regulatory change whenever its **relative marginal cost falls**:

$$\frac{\psi^R_\omega}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

even if its own marginal cost rises, $\psi^R_\omega > \psi_\omega$.

Whether this happens — and for which firms — depends on how the regulation deforms the cost distribution.

Regulations

Suppose marginal costs come from a family of continuous parametric CDFs, $\psi \sim G \in \mathcal{G}$.

Def. Regulation:

A regulation changes marginal costs to $\psi^R \sim G^R$, with $G, G^R \in \mathcal{G}$.

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Why parameterize? It lets us classify regulations by how they deform the distribution: **scale**, **location**, or **shape**.

Shape changes raise costs for everyone but have a vanishing burden on the most productive.

E.g., in Pareto costs: $G(\psi) = 1 - \left(\frac{\psi_0}{\psi}\right)^{\kappa}$, κ and ψ_0 are the shape and scale parameters, respectively.

Shape Regulations

Thm 1:

A regulation that raises costs through the shape of the distribution ($\psi_\omega^R > \psi_\omega \ \forall \omega$, via κ) benefits the most productive firms.

→ Minimal statement for non-parametric ψ, no dominance between ψ and ψ^R.

→ Adding choke prices amplifies benefits for surviving firms.

Proof of Thm 1

Proof: The most productive firms always benefit.

- Since $\psi_\omega^R > \psi_\omega \ \forall \omega \in (0,1)$, $A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$.
- By continuity of $\mathcal{G}$ and support-invariance of κ:

$$\lim_{\omega \downarrow 0} \frac{\psi_\omega^R}{\psi_\omega} = 1$$

This implies there exists $\varepsilon > 0$ so that $\forall \omega \in (0, \varepsilon)$:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

■

Scale Regulations (boundary: neutrality)

Thm 2:

A proportional cost increase ($\psi_\omega^R = r\psi_\omega$, $r > 1$) leaves all firms unaffected.

By homogeneity of $A(\cdot)$:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} = \frac{r\psi_\omega}{A(r\boldsymbol{\mu\psi})} = \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

Proportional compliance wedges (ad valorem taxes, uniform input price increases) generate no rents from $A(\cdot)$.

Location Regulations (boundary: harms top)

Thm 3:

A flat cost increment ($\psi_\omega^R = \psi_\omega + a$, $a > 0$) harms the most productive firms.

A flat increment a is proportionally larger for low-cost firms. HSA regularity and IVT imply there exists a cutoff $\varepsilon \in (0, 1)$ such that:

$$\frac{A(\boldsymbol{\mu}(\boldsymbol{\psi} + a))}{A(\boldsymbol{\mu}\boldsymbol{\psi})} = \frac{\psi_\varepsilon + a}{\psi_\varepsilon}$$

For every $\omega \in (0, \varepsilon)$ the proportional increase is greater, so their relative marginal cost rises. Examples: per-unit licensing fees, flat compliance costs, mandatory lump-sum reporting requirements.

→ Full proof (existence of ε).

Illustration

Relative Cost Under CES-Pareto

Under CES demand with Pareto costs $\psi \sim \text{Pa}(\psi_0, \kappa)$:

$$A(\mu\boldsymbol{\psi}) = \left[\int_0^1 \left(\frac{\sigma}{\sigma-1}\psi_\gamma\right)^{1-\sigma} d\gamma\right]^{\frac{1}{1-\sigma}}$$

The relative marginal cost for firm ω is:

$$\frac{\psi_\omega}{\mu A(\boldsymbol{\psi})} = \frac{(1-\omega)^{-\frac{1}{\kappa}}}{\mu\left[\frac{\kappa}{\kappa+\sigma-1}\right]^{\frac{1}{1-\sigma}}}$$

Who Benefits?

Suppose a regulation marginally decreases κ, so $\psi_\omega^R > \psi_\omega \ \forall \omega$. By Thm 1, every firm in $(0, \varepsilon)$ benefits. A firm ω benefits if:

$$\frac{\partial \ln(\psi_\omega / A(\boldsymbol{\psi}))}{\partial \kappa} > 0$$

The cutoff is:

$$\varepsilon = 1 - \exp\left(-\frac{\kappa}{\kappa + \sigma - 1}\right)$$

A larger elasticity of substitution or a less dispersed cost distribution ($\uparrow \sigma$ or $\uparrow \kappa$) widens the set of benefiting firms and increases their gains.

Application: Minimum Wage

The Empirical Puzzle

Robust pattern across countries and settings after $\uparrow w_{\min}$:

- Workers reallocate from low- to high-productivity firms.
  Rao and Risch (2026); Dustmann et al. (2022); Engbom and Moser (2022); Haelbig et al. (2023).
- High-productivity firms expand, gain market share, and appear more productive. Low-productivity firms contract.
  Rao and Risch (2026); Mayneris et al. (2018); Draca et al. (2011); Luca and Luca (2018).

The puzzle:

Why do firms already paying **above** the minimum wage expand and become more productive?

Standard Explanations

The literature attributes these patterns to:

- **Monopsony:** the minimum wage cuts markdown of low productivity firms, they ration employment and workers reallocate up the productivity ladder
- **Search frictions:** the floor raises costs for low-productivity firms; they post fewer vacancies, reducing congestion; high-productivity firms fill their openings more easily and expand.
- **Efficiency wages:** higher wages reduce turnover and shirking; productivity rises directly.
- **Managerial slack:** cost pressure forces reorganization; firms improve or exit.

All fundamentally labor market mechanisms.

Why Labor Market Stories Fall Short (1/2)

Monopsony & Search Frictions (Berger et al. 2025; Dustmann et al. 2022; Haanwinckel 2025; Engbom and Moser 2022; Drechsel-Grau 2023)

Both require market frictions that are hardest to sustain in low-skill labor markets:

- **Thick markets:** many similar employers, substitutable workers — the "few alternatives" premise is implausible.
- **Standardized jobs:** low specialization $\Rightarrow$ fast, cheap matching $\Rightarrow$ search frictions are mechanically small.
- **Empirically:** near-complete price pass-through (Allegretto and Reich 2018; Harasztosi and Lindner 2019); price-to-wage ratios close to 1 in US restaurants (Aaronson and French 2007) undercut the monopsony premise.

Low-skill labor is precisely where competitive markets work best.

Why Labor Market Stories Fall Short (2/2)

Managerial Slack & Efficiency Wages (Hau et al. 2020; Mayneris et al. 2018; Riley and Rosazza Bondibene 2017)

- **Issue 1: Why wait for a regulatory shock?**
  If reorganization raises productivity (slack), firms would already adopt it. If paying above market improves effort (efficiency wages), firms can already do so. Neither mechanism requires a mandate.

- **Issue 2: One-sided.**
  Both predict improvements only for firms directly hit by the floor — firms whose workers received a pay increase. Yet firms already paying well **above** the floor expand and gain. Draca et al. 2011; Haelbig et al. 2023

- **Issue 3: Spillovers.**
  MW directly affects only low-wage workers. Firm-wide productivity gains require large positive complementarities across the entire workforce.

Economy-wide Extension

The minimum wage is uniform across industries — a single-industry model misses cross-industry demand substitution. We embed the mechanism in a nested CES economy: within-industry elasticity σ_g, across-industry elasticity σ_0.

Profit of firm ω in industry g:

$$\pi_\omega^g = \frac{YS_g}{\sigma_g}\left(\frac{\sigma_g}{\sigma_g - 1} \cdot \frac{\psi_\omega}{P_g}\right)^{1-\sigma_g}$$

where S_g is total expenditure on industry g, determined by P_g relative to other industries' price indices.

Key property: maximizing profit over prices is equivalent to maximizing over the relative marginal cost ψ_ω/P_g. Profit is decreasing in this ratio.

Minimum Wage as Shape Regulation

High-productivity firms are already paying above the floor — their direct cost increase is negligible; P_g rises because low-productivity rivals bear the burden.

Within the industry: top firms' relative marginal cost ψ_ω / P_g falls, their within-industry share and profit rise.

Across industries: P_g rising redirects some spending elsewhere. Under $\sigma_g > \sigma_0$, the within-industry gain dominates the cross-industry outflow — even if no other industry's price rises.

Theorem 1.1:

Under $\sigma_g > \sigma_0$, the most productive firms in **every** industry gain in profit after an economy-wide shape-changing regulation.

$\sigma_g > \sigma_0$ is the standard assumption — goods within an industry are closer substitutes than goods across industries.

Proof of Theorem 1.1

Three channels (last two from decomposing $d \ln S_g$):

$$d \ln \pi_\omega^g = \underbrace{(\sigma_g - 1)(d \ln P_g - d \ln \psi_\omega)}_{\text{within-industry position}} - \underbrace{(\sigma_0 - 1)(1 - S_g)\, d \ln P_g}_{\text{outflows}} + \underbrace{(\sigma_0 - 1)(1 - S_g)\, d \ln \bar{P}_{-g}}_{\text{inflows}}$$

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Proof of Theorem 1.1

Three channels:

$$d \ln \pi_\omega^g = \left[(\sigma_g - 1) - (\sigma_0 - 1)(1 - S_g)\right] d \ln P_g > 0$$

- **Left-tail convergence:** $d \ln \psi_\omega \to 0$ — own cost increase vanishes.
- **Worst case:** $d \ln \bar{P}_{-g} = 0$ — no inflows (single-industry shock).
- **Net condition:** $\sigma_g - 1 > (\sigma_0 - 1)(1 - S_g)$. Sufficient: $\sigma_g > \sigma_0$. ■

Testable Predictions

The mechanism predicts:

- **Relative prices:** least productive firms raise prices more after $\uparrow w_{\min}$; most productive raise prices less.
- **Market shares:** productive firms gain market share within the industry, even after accounting for exit; unproductive firms lose.
- **Employment:** follows output — productive firms hire more because demand shifted, not from labor market effects.

All predictions require within-industry data on firm-level prices and market shares — the level of analysis used in none of the existing papers.

Conclusion

- **Mechanism:** firm outcomes are determined by **relative** marginal cost. Regulation can shift the price aggregator enough to increase market share and profits for the most productive firms.
- **Classification:** scale regulation is **neutral**; location regulation **harms** the most productive; only shape regulation **benefits** them.
- **Application:** the minimum wage is a shape-type regulation. Firms above the floor expand and **appear** more productive: demand shifts toward them, raising revenues and profits — rents, not efficiency. Under $\sigma_g > \sigma_0$, the result holds economy-wide.
- **Inequality:** the MW is designed to reduce labor income inequality. But the product market channel redistributes profit toward high-productivity firms — increasing capital income inequality. The net effect on overall inequality is ambiguous.

Appendix Index

- Detailed Literature Review.
- Minimum Wage Literature.
- Thm 1: Minimal statement.
- Choke Prices.
- Proof of Thm 3.
- Why measured productivity rises.
- References.

Literature Review

1. Regulation as a Traded Good / Regulator as Agent.

- **Core idea:** regulation is endogenous; organized interests demand it; policymakers supply it.
- **Seminal:** Tullock (1967), Stigler (1971), Peltzman (1976), Becker (1983), Laffont and Tirole (1993).
- **Common-agency benchmark:** Grossman and Helpman (1994).
- **Heterogeneous lobbying:** Bombardini (2008).
- **Survey:** Dal Bó (2006); Bombardini and Trebbi (2011, 2020).
- **What we add:** this paper takes the regulation as given and asks who benefits within the industry. A new **within-industry conflict object**:

$$\text{winners/losers determined by } \frac{\psi_\omega}{A(\mu\psi)} \text{ rather than by } \psi_\omega.$$

In canonical political-economy settings, incentives within an industry are aligned; the conflict is across industries. This paper shows within-industry conflict arises even from a rule that applies to all.

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2. Raising Rivals' Costs (IO/PE).

Early evidence: Marvel (1977): first enforced social policy (child labor laws) as capture.
Canonical modeling:

- **Formal RRC:** Salop and Scheffman (1983, 1987): regulations, standards, input contracts. Finite-firm oligopoly; strategic interaction; partial equilibrium.
- Entry deterrence: Ordover et al. (1990). Raising rivals' fixed costs: Hviid and Olczak (2016).

How our mechanism differs:

- RRC advantage requires asymmetric incidence: exemptions, enforcement differences, or shocks that hit rivals by design.
- Here: even industry-wide marginal-cost regulation creates winners via shifts in $A(\cdot)$. The heterogeneous effective burden arises endogenously from productivity differences under the same rule.

Related: Kennard (2020) (heterogeneous compliance in environmental settings); Muehlegger and Sweeney (2022) (own vs rivals' cost pass-through).

←

3. Firm Size and Capture (Applied/Macro).

Modern theory: large firms favor stricter regulation

- Macedoni and Weinberger (2024) (CES, Pareto productivity): large firms can favor regulation raising **fixed costs** — induces exit of small rivals, raises markups.
- Their logic implies **neutrality of marginal-cost regulation** (scale changes) — consistent with Thm 2.
- **Our distinction:** variable-cost regulation matters when it changes the **shape** of the cost distribution (Thm 1).

Consistent evidence:

- Crane and Koch (2025): valuation/markup/entry patterns for connected firms in regulated sectors.
- Fan and Zhou (2023): markups and political connections.
- Huneeus and Kim (2024): lobbying and firm size in a macro GE setting.

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Minimum Wage Literature

4. Minimum Wage and Firm Outcomes (Labor).

Reallocation evidence:

- Dustmann et al. (2022): worker reallocation toward high-productivity firms after German MW introduction.
- Engbom and Moser (2022): MW compresses earnings distribution; high-productivity firms gain workers.
- Haelbig et al. (2023): firm-level evidence, gains concentrated at top of productivity distribution.
- Rao and Risch (2026): firm expansion above the floor; primary product-market-mechanism paper.

Product market frameworks (and their limits):

- Williamson (1968); Haucap et al. (2001): early RRC logic applied to MW. Homogeneous goods, Cournot.
- Harasztosi and Lindner (2019): imposes proportional cost shock (R-cic) — neutral with respect to relative prices (Thm 2). No within-industry redistribution.
- Luca and Luca (2018): restaurant-sector intra-industry evidence consistent with RRC; explicit Williamson (1968) citation.

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Thm 1: Minimal

Thm 1: Minimal.

If the following two conditions hold:

1. Strict index ordering:
$$A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$$
2. Left-tail convergence to 1:
$$\lim_{\omega \downarrow 0} \frac{\psi_\omega^R}{\psi_\omega} = 1$$

Then there exists $\varepsilon > 0$ so that $\forall \omega \in (0, \varepsilon)$:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

←

Choke Prices

Choke Prices

Choke prices are not required for capture via $A(\cdot)$; they act as an amplifier by inducing exit.

As in Matsuyama and Ushchev (2020), let choke prices be $\bar{p}_\omega = \bar{z}_\omega A(\mathbf{p}) > 0$ and the HSA price aggregator $\boldsymbol{A(\cdot)}$ **implicitly defined** by:

$$\int_\Omega s_\omega \left(\frac{\mu_\omega \psi_\omega}{A(\boldsymbol{\mu\psi})} \right) \partial\omega = 1 \text{ for all possible } \boldsymbol{\mu\psi}$$

Without choke prices, $A(\cdot)$ is defined for all possible $\boldsymbol{\mu\psi}$. With binding choke prices, exit changes the measure Ω, implicitly changing $A(\cdot)$.

Choke Prices

Suppose R-cis induces a positive measure of firms to exit, $|\Omega| > |\Omega^R|$. Then $A^R(\cdot)$ is implicitly defined over Ω^R. Since the share functions $s_\omega(\cdot)$ do not change and the domain shrinks, it must be that $A^R(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi}^R)$.

Cor 1: Exit as an amplifier.

When choke prices bind, the most productive surviving firms reap strictly larger benefits from R-cis.

Proof. From Thm 1, $A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$ and $\exists\, \varepsilon > 0$ s.t. every $\omega \in (0, \varepsilon)$ benefits. From above, $A^R(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi}^R)$, so for all $\omega \in (0, \varepsilon) \cap \Omega^R$:

$$\frac{\psi_\omega^R}{A^R(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)}$$

■

Exit is not necessary but amplifies a mechanism that already operates through $A(\cdot)$.

←

Proof. Thm 3

Proof of Thm 3 (existence of ε)

Thm 3: The most productive firms are negatively affected by R-cil.

Let $\psi_\omega = G^{-1}(\omega)$ be continuous, strictly increasing, $\psi_\omega \geq \psi_0 > 0$, $\psi_\omega \to \infty$ as $\omega \uparrow 1$.
Let $A(\cdot)$ be monotone and homogeneous of degree 1.
Define $\rho(a) \equiv A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}))/A(\boldsymbol{\mu}\boldsymbol{\psi})$, $a > 0$.
Step 1 (bounds on $\rho(a)$). Monotonicity implies $\rho(a) > 1$. Since $\psi_\omega \geq \psi_0$,

$$\psi_\omega + a \leq \left(1 + \frac{a}{\psi_0}\right)\psi_\omega \ \forall\omega \ \Rightarrow \ \rho(a) \in \left(1,\, 1 + \frac{a}{\psi_0}\right].$$

Proof of Thm 3 (existence of ε)

Step 2 (IVT). Let $g(\omega) = 1 + a/\psi_\omega$. Then g is continuous, strictly decreasing, with

$$\lim_{\omega \downarrow 0} g(\omega) = 1 + \tfrac{a}{\psi_0}, \qquad \lim_{\omega \uparrow 1} g(\omega) = 1.$$

Since $\rho(a) \in (1,\, 1 + a/\psi_0]$, IVT gives a unique $\varepsilon \in (0, 1)$ with $g(\varepsilon) = \rho(a)$.
For $\omega < \varepsilon$, $g(\omega) > g(\varepsilon)$, so $(\psi_\omega + a)/A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1})) > \psi_\omega/A(\boldsymbol{\mu}\boldsymbol{\psi})$: top firms are harmed. ■

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Why Measured Productivity Rises

Why Measured Productivity Rises

- **Labor productivity** $\equiv$ (remuneration + **profits**) / employment.
  Riley and Rosazza Bondibene (2017). Profit is included.
- High-productivity firms gain market share $\Rightarrow$ revenues rise $\Rightarrow$ profits rise $\Rightarrow$ measured productivity rises.
- **This is not** a genuine efficiency gain — it is a **competitive position** gain.
- Same logic applies to TFP: higher output at similar inputs $\Rightarrow$ positive Solow residual. Scale effects and markup changes contaminate the residual.

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